Lab 8 — F# Code

Question 2: Database CRUD — Relational (SQLite) and NoSQL (MongoDB) (Exp 15)

Setup:

```
dotnet new console -lang F# -o Lab8Q2
cd Lab8Q2
dotnet add package Microsoft.Data.Sqlite
dotnet add package MongoDB.Driver
```

MongoDB must be running locally on port 27017 (e.g. `docker run -d -p 27017:27017 mongo`).

Part 1 — SqliteCrud.fs

```
module SqliteCrud

open Microsoft.Data.Sqlite

let exec (conn: SqliteConnection) (sql: string) (ps: (string * obj) list) =
    use cmd = conn.CreateCommand()
    cmd.CommandText <- sql
    for (k, v) in ps do
        cmd.Parameters.AddWithValue(k, v) |> ignore
    cmd.ExecuteNonQuery() |> ignore

let printAll (conn: SqliteConnection) (label: string) =
    printfn "--- %s ---" label
    use cmd = conn.CreateCommand()
    cmd.CommandText <- "SELECT Id, Name, Age FROM Student"
    use reader = cmd.ExecuteReader()
    while reader.Read() do
        printfn "Id=%d, Name=%s, Age=%d"
            (reader.GetInt32 0) (reader.GetString 1) (reader.GetInt32 2)

let run () =
    printfn "===== SQLite CRUD ====="
    use conn = new SqliteConnection("Data Source=students.db")
    conn.Open()

    exec conn "DROP TABLE IF EXISTS Student" []
    exec conn "CREATE TABLE Student (Id INTEGER PRIMARY KEY, Name TEXT, Age INTEGER)" []

    // CREATE
    exec conn "INSERT INTO Student (Id, Name, Age) VALUES (@id, @name, @age)"
         [ "@id", box 1; "@name", box "Alice"; "@age", box 20 ]
    exec conn "INSERT INTO Student (Id, Name, Age) VALUES (@id, @name, @age)"
         [ "@id", box 2; "@name", box "Bob"; "@age", box 22 ]
    printAll conn "After CREATE"

    // READ
    printAll conn "READ"

    // UPDATE
    exec conn "UPDATE Student SET Age = @age WHERE Id = @id"
         [ "@age", box 21; "@id", box 1 ]
    printAll conn "After UPDATE (Alice age -> 21)"

    // DELETE
    exec conn "DELETE FROM Student WHERE Id = @id" [ "@id", box 2 ]
    printAll conn "After DELETE (Bob removed)"
```

Part 2 — MongoCrud.fs

```
module MongoCrud

open MongoDB.Bson
open MongoDB.Driver

let run () =
    printfn "\n===== MongoDB CRUD ====="
    let client = MongoClient("mongodb://localhost:27017")
    let db = client.GetDatabase("labdb")
    let col = db.GetCollection<BsonDocument>("students")

    // start clean
    col.DeleteMany(FilterDefinition<BsonDocument>.Empty) |> ignore

    let printAll (label: string) =
        printfn "--- %s ---" label
        col.Find(FilterDefinition<BsonDocument>.Empty).ToList()
        |> Seq.iter (fun d ->
            printfn "Name=%s, Age=%d" (d.["name"].AsString) (d.["age"].AsInt32))

    // CREATE
    col.InsertOne(BsonDocument([ BsonElement("name", BsonString "Alice")
                                 BsonElement("age", BsonInt32 20) ]))
    col.InsertOne(BsonDocument([ BsonElement("name", BsonString "Bob")
                                 BsonElement("age", BsonInt32 22) ]))
    printAll "After CREATE"

    // READ
    printAll "READ"

    // UPDATE
    let filterAlice = Builders<BsonDocument>.Filter.Eq("name", "Alice")
    let update = Builders<BsonDocument>.Update.Set("age", 21)
    col.UpdateOne(filterAlice, update) |> ignore
    printAll "After UPDATE (Alice age -> 21)"

    // DELETE
    let filterBob = Builders<BsonDocument>.Filter.Eq("name", "Bob")
    col.DeleteOne(filterBob) |> ignore
    printAll "After DELETE (Bob removed)"
```

Program.fs

```
[<EntryPoint>]
let main _ =
    SqliteCrud.run ()
    MongoCrud.run ()
    0
```

Lab8Q2.fsproj (file order matters in F#)

```
<ItemGroup>
  <Compile Include="SqliteCrud.fs" />
  <Compile Include="MongoCrud.fs" />
  <Compile Include="Program.fs" />
</ItemGroup>
```